

NASA C-MAPSS FD001 portfolio model built entirely in Excel. The project combines Power Query, Power Pivot/DAX, reliability analysis, leakage-safe validation, interpretable Remaining Useful Life regression, empirical uncertainty adjustment and Solver-based maintenance scheduling.

FLEET RELIABILITY

Median life 199 cycles

HELD-OUT VALIDATION

MAE 24.94 | R2 0.746

NASA FINAL TEST

MAE 22.85 | RMSE 28.44

CURRENT FLEET STATUS

54 Critical | 15 Warning  
31 Normal



Source: NASA C-MAPSS FD001. Maintenance thresholds are portfolio assumptions, not approved aviation maintenance limits.

Development workflow

1. Split by engine

80 Development engines / 20 held-out Validation engines.

2. Screen sensors

All 21 sensors assessed using Development-only variability, correlation and degradation shift.

3. Build Health Score

Selected sensors: S11, S04, S12, S07 and S15. Equal-weight degradation score, clamped to 0-100.

4. Compare RUL models

Naive baseline -> Cycle + 5 sensors -> Cycle + Cycle^2 + 5 sensors.

5. Lock then test

Winning specification locked before final refit on all training engines and evaluation on 100 unseen NASA test engines.

Validation and final test evidence

| Internal Validation (Held-Out Engines)                   |       |       |       |                                         |                                          |                                                                                 |  |  |  |                         |       |  |  |  |
|----------------------------------------------------------|-------|-------|-------|-----------------------------------------|------------------------------------------|---------------------------------------------------------------------------------|--|--|--|-------------------------|-------|--|--|--|
| Metric                                                   |       |       |       |                                         | Model Selection                          |                                                                                 |  |  |  | NASA Final Test Results |       |  |  |  |
| Naive Baseline Sensor Model Cycle^2 Model Interpretation |       |       |       |                                         | Selected Model                           |                                                                                 |  |  |  | Metric                  |       |  |  |  |
| MAE                                                      | 28.65 | 27.58 | 24.94 | Lower is better                         | Cycle + Cycle^2 + 5 Standardized Sensors |                                                                                 |  |  |  | MAE                     | 22.85 |  |  |  |
| RMSE                                                     | 36.43 | 34.58 | 31.88 | Lower is better; penalises large errors | Reason                                   | Lowest validation MAE/RMSE, highest predictive R^2, and lower bias than Model 1 |  |  |  | RMSE                    | 32.85 |  |  |  |
| Bias                                                     | 4.11  | 11.81 | 9.75  | Closer to zero is better                | Status                                   | LOCKED BEFORE NASA TEST                                                         |  |  |  | RMSE                    | 32.85 |  |  |  |
| Predictive R^2                                           | 0.988 | 0.701 | 0.788 | Higher is better                        |                                          |                                                                                 |  |  |  | Bias                    | 9.71  |  |  |  |
|                                                          |       |       |       |                                         |                                          |                                                                                 |  |  |  | Predictive R^2          | 0.552 |  |  |  |
|                                                          |       |       |       |                                         |                                          |                                                                                 |  |  |  | Test Engines            | 100   |  |  |  |
| Validation Observations                                  |       |       |       |                                         |                                          |                                                                                 |  |  |  |                         |       |  |  |  |
| Validation Engines                                       |       |       |       |                                         |                                          |                                                                                 |  |  |  |                         |       |  |  |  |
| P90 Validation Abs Error (Selected Model)                |       |       |       |                                         |                                          |                                                                                 |  |  |  |                         |       |  |  |  |
| 55.90                                                    |       |       |       |                                         |                                          |                                                                                 |  |  |  |                         |       |  |  |  |

| NASA Final Test: 100 Unseen Engines |               |           |           |           |           |           |              |              |              |              |              |              |               |             |              |
|-------------------------------------|---------------|-----------|-----------|-----------|-----------|-----------|--------------|--------------|--------------|--------------|--------------|--------------|---------------|-------------|--------------|
| Unit ID                             | Current Cycle | Sensor_11 | Sensor_04 | Sensor_12 | Sensor_07 | Sensor_15 | Z_S11        | Z_S04        | Z_S12        | Z_S07        | Z_S15        | Cycle2       | Predicted_RUL | Actual_RUL  |              |
| 1                                   | 21            | 47.25     | 1289.81   | 521.79    | 1054.42   | 0.4023    | -1.185642454 | 1.189022942  | -1.059720813 | 881          | 187.2381876  | 112          | 75.03819754   | 75.03819754 |              |
| 2                                   | 49            | 47.67     | 1410.85   | 521.74    | 1053.52   | 0.4005    | 0.407358416  | 0.310917957  | 0.443779645  | 0.177058767  | 0.237245017  | 2451         | 131.8038348   | 131.8038348 |              |
| 3                                   | 126           | 47.89     | 1418.89   | 520.83    | 1052.93   | 0.4119    | 1.08917951   | 0.77598919   | 0.918913812  | -0.00640309  | 1.6878       | 42.28317901  | 46            | 6.11420993  |              |
| 4                                   | 126           | 47.85     | 1406.88   | 521.88    | 1052.64   | 0.48634   | 0.407470032  | -0.228192632 | 0.335372729  | 0.588796348  | 1.1336       | 80.426173763 | 82            | 6.425737927 |              |
| 5                                   | 89            | 47.48     | 1419.38   | 521       | 1053.28   | 0.4390    | 0.303801085  | 1.15830088   | -0.00586731  | -0.087800125 | 0.188527953  | 8854         | 85.10589167   | 91          | 2.105891616  |
| 6                                   | 129           | 47.57     | 1402.89   | 521.98    | 1053.79   | 0.4852    | -0.118899427 | -0.892222244 | 0.158810328  | 0.802221983  | 0.001482028  | 11025        | 156.9448862   | 93          | 11.94488619  |
| 7                                   | 100           | 47.32     | 1413.57   | 522.52    | 1053.42   | 0.4023    | -0.836674054 | 0.815100754  | 1.22910421   | 0.00877218   | -1.048074578 | 35808        | 87.83706832   | 91          | 1.807068515  |
| 8                                   | 186           | 47.89     | 1413.89   | 520.63    | 1053.61   | 0.4744    | 0.183820986  | 0.50095891   | 1.18783813   | 0.180761534  | 0.600022929  | 2795         | 87.34873348   | 95          | 37.88328693  |
| 9                                   | 56            | 47.58     | 1410.27   | 521.85    | 1052.98   | 0.4293    | 0.145202086  | 0.188181225  | 0.20988359   | -0.88937889  | -0.242526337 | 32026        | 127.808865    | 111         | 18.80886497  |
| 10                                  | 182           | 47.58     | 1388.89   | 521.82    | 1052.85   | 0.4184    | 0.182830885  | -1.145478383 | 0.511871817  | -0.515328672 | -0.004815776 | 36864        | 78.80214431   | 86          | -18.0885558  |
| 11                                  | 83            | 47.87     | 1414.22   | 520.78    | 1052.8    | 0.4864    | 0.482328848  | 0.587318124  | -0.93113583  | -0.84741472  | 0.682701989  | 8888         | 80.04424278   | 97          | -0.855767388 |
| 12                                  | 217           | 47.37     | 1409.66   | 522.03    | 1053.38   | 0.4482    | -0.048889419 | 0.078483429  | 0.03812328   | -0.009717235 | 0.161425862  | 47889        | 78.21281118   | 128         | -45.78861882 |
| 13                                  | 186           | 47.52     | 1414.61   | 521.5     | 1053.28   | 0.4258    | -0.078925488 | 0.874818181  | -0.133688119 | -0.00988888  | 0.277801878  | 38525        | 85.58486813   | 95          | -38.44835987 |
| 14                                  | 86            | 47.57     | 1408.35   | 520.88    | 1053.75   | 0.4233    | -1.07819418  | -0.048892714 | 1.521888927  | 0.421781927  | -0.940881348 | 2118         | 137.8186386   | 127         | 30.97869357  |
| 15                                  | 76            | 47.25     | 1400.81   | 522.12    | 1054.08   | 0.4067    | -0.904882487 | 0.8578274    | 0.804782196  | -0.840088549 | 0.779        | 147.8413111  | 83            | 64.84513108 |              |
| 16                                  | 113           | 47.37     | 1401      | 521.77    | 1054.8    | 0.4289    | -0.048889415 | -0.881470986 | 0.493398484  | 1.882271327  | -0.33804888  | 1288         | 117.7670378   | 84          | 33.70702717  |
| 17                                  | 186           | 47.58     | 1412.84   | 521.52    | 1053.49   | 0.4310    | 0.033881288  | 0.008884     | 0.144388919  | 0.138188019  | 0.371188817  | 25228        | 73.14127671   | 80          | 23.14127671  |
| 18                                  | 133           | 47.7      | 1418.18   | 521.06    | 1053.56   | 0.4418    | 0.04681351   | 1.138393264  | -0.47934888  | -0.009717235 | -0.00014379  | 17889        | 61.1023774    | 28          | 38.16202774  |
| 19                                  | 135           | 47.35     | 1408.32   | 521.48    | 1053.87   | 0.4788    | 0.715751771  | 0.280400673  | 0.280208811  | 0.587488784  | 0.82889882   | 18225        | 85.84282888   | 87          | 5.18238888   |
| 20                                  | 184           | 48.12     | 1419.8    | 520.64    | 1052.18   | 0.489     | 2.39481088   | 1.188593726  | 1.54894888   | 1.30588421   | 0.196574652  | 2388         | 23.4320528    | 18          | 7.82581808   |
| 21                                  | 148           | 47.48     | 1403.77   | 522.02    | 1053.88   | 0.441     | -0.305801088 | -0.937318866 | 0.333353886  | 0.380500517  | -0.000484747 | 71884        | 83.81815648   | 57          | 36.41850644  |
| 22                                  | 38            | 47.38     | 1410.18   | 521.77    | 1054.04   | 0.4163    | 0.818931043  | 0.138221324  | 0.483288488  | 0.719589187  | 0.74288108   | 1521         | 181.280188    | 111         | 50.280188    |
| 23                                  | 130           | 47.22     | 1403.86   | 521.83    | 1054.15   | 0.4349    | -1.202483332 | -0.852888567 | 0.84745537   | 0.88384888   | -0.45820508  | 18806        | 112.3042307   | 113         | -0.89884275  |
| 24                                  | 146           | 47.3      | 1415.15   | 520.88    | 1053.78   | 0.4785    | 0.566843541  | -0.00586538  | 0.484514475  | 0.488810538  | 0.488810538  | 2406         | 58.4187358    | 30          | 30.8187358   |
| 25                                  | 48            | 47.38     | 1403.1    | 521.75    | 1053.55   | 0.4422    | -0.00342888  | -0.648134411 | 0.418278002  | 0.200180058  | 0.200180058  | 2354         | 131.3804881   | 145         | 0.3804881    |
| 26                                  | 78            | 47.4      | 1402.4    | 521.83    | 1054.45   | 0.4331    | -0.528948832 | -0.728488862 | 0.703288832  | 1.222787713  | -0.24118115  | 8778         | 135.4727747   | 118         | 16.47277489  |
| 27                                  | 142           | 47.48     | 1407.16   | 521.64    | 1053.3    | 0.44      | -0.178782221 | -0.187278111 | 0.37437888   | -0.78081888  | 0.1808       | 88.2882481   | 88            | 23.2882481  |              |
| 28                                  | 146           | 47.58     | 1408.18   | 521.81    | 1053.18   | 0.4488    | 0.18081888   | 0.18081888   | 0.18081888   | 0.18081888   | 0.18081888   | 87           | -18.18081888  | 18          | 18.18081888  |

Selected model

Cycle + Cycle^2 + five standardised sensors

Validation MAE

24.94

Test RMSE

28.44

Validation RMSE

31.88

Test bias

+9.71

Validation R2

0.746

Test R2

0.532

Test MAE

22.85

Empirical uncertainty

P90 held-out absolute error = 55.90 cycles.

Conservative RUL = MAX(0, Predicted RUL - 55.90).

This is an empirical validation error band, not a statistical confidence interval.

Interactive engine condition monitoring

Select any unseen test engine to inspect current cycle, predicted RUL, Conservative RUL, Health Score, maintenance status and full sensor / health / RUL trajectories.



Solver-based maintenance allocation

The 20 highest-priority engines are allocated across four maintenance windows with binary decision variables, one assignment per engine and five-engine capacity per window. Solver minimises urgency-weighted delay.

